A Counterexample to ℓ_∞-Stability of Quasi-ACK Structure

Research report

June 23, 2026

Abstract

Choi and Jung asked whether arbitrary ℓ_∞-sums of Banach spaces with quasi-ACK structure again have quasi-ACK structure, and whether the analogous assertion holds for property quasi-β. We first verify the supplied partial result: a uniform bound below one on the active ρ-values upgrades each quasi-ACK witness to an ordinary ACK_r witness, after which the standard coordinate argument applies. We then give a countable real counterexample to the unrestricted question. The summands are two-dimensional regular-polygon spaces X_n, each having property β (indeed $\mathrm{ACK}_{\cos(\pi/m_n)}$ structure), where $m_n \to \infty$. Nevertheless,

$$Z = \Big(\bigoplus_{n=1}^{\infty} X_n\Big)_{\ell_\infty}$$

does not have quasi-ACK structure. The proof identifies every extreme point of B_{Z^*} with an extreme functional in an ultrafilter fiber. Free ultrafilters see the Euclidean limit of the polygonal norms. A local consequence of the quasi-ACK axioms is then incompatible with the round geometry of that limit fiber. Consequently, arbitrary ℓ_∞-sum stability fails for both quasi-ACK structure and property quasi-β over the real field.

1 The question and the conclusion

In Remarks 3.4 and 4.8 of Choi–Jung [2], the following question is left open: if every member of a family $(X_i)_{i\in I}$ has quasi-ACK structure, must $(\bigoplus_{i\in I} X_i)_{\ell_\infty}$ have quasi-ACK structure? The paper also asks the same question for property quasi-β. Finite ℓ_∞-sums cause no difficulty, since they are finite c_0-sums; the issue is genuinely infinite.

Corollary 3.3. *Let Z be a Banach space with quasi-*ACK *structure.*

(a) *If K is a compact Hausdorff space, then $C(K, Z)$ has quasi-*ACK *structure.*

(b) *If Ω is a completely regular Hausdorff space and if, in addition, Z is finite dimensional, then $C_b(\Omega, Z)$ has quasi-*ACK *structure.*

(c) *$c_0(Z)$ has quasi-*ACK *structure.*

Proof. (a) is an obvious consequence of Theorem 3.1. For the item (b), note that $C_b(\Omega, Z)$ can be isometrically identified with $C(\beta\Omega, Z)$, where $\beta\Omega$ is the Stone-Čech compactification of Ω. (c) follows from the observation that $c_0(Z)$ can be viewed as a closed subspace of $C(\beta\mathbb{N}, Z)$. □

Remark 3.4. We do not know if the class of Banach spaces with quasi-ACK structure is stable under ℓ_∞-sum operations in general. As far as we know, it is even unknown if property quasi-β is stable under ℓ_∞-sums or not. Nevertheless, the assertion (b) of Corollary 3.3 yields that $\ell_\infty(Z)$ has quasi-ACK structure whenever Z is a finite dimensional Banach space with quasi-ACK structure.

In fact, Corollary 3.3 enables us to construct a non-trivial Banach space which is of quasi-ACK structure, but not ACK$_\rho$ structure or property quasi-β. We first need the following technical lemma.

Lemma 3.5. *Let K be a compact Hausdorff space and Z be a Banach space. If $C(K, Z)$ has property* B*, then $C(K)$ has property* B.

Proof. Let X be a Banach space and $T \in \mathcal{L}(X, C(K))$ with $\|T\| = 1$. Fix $z_0 \in S_Z$ and define $\widetilde{T} \in \mathcal{L}(X, C(K, Z))$ by

The answer for arbitrary families is negative.

Theorem 1.1 (Main result)**.** *There is a sequence $(X_n)_{n\geq 1}$ of two-dimensional real Banach spaces such that every X_n has property β, but*

$$\Big(\bigoplus_{n=1}^{\infty} X_n\Big)_{\ell_\infty}$$

*does not have quasi-*ACK *structure. Hence neither quasi-*ACK *structure nor property quasi-β is stable under arbitrary ℓ_∞-sums.*

The proof is self-contained apart from standard forms of Krein–Milman and Milman's converse. Since the argument appears not to have been published in the literature located for this report, it should be regarded as a new, unrefereed proof and checked independently before citation.

2 Definitions

All spaces in the counterexample are real. A set $\Gamma \subset B_{X^*}$ is 1-*norming* if

$$\|x\| = \sup_{x^*\in\Gamma} |x^*(x)| \qquad (x \in X).$$

For a subset A of a vector space, $\mathrm{aco}(A)$ denotes its absolutely convex hull.

Definition 2.1 (Choi–Jung)**.** A Banach space X has *quasi-*ACK *structure* if there are a 1-norming set $\Gamma \subset B_{X^*}$ and a function $\rho : \Gamma \to [0, 1)$ such that, for every $e^* \in \mathrm{Ext}(B_{X^*})$, there is a set $\Gamma_{e^*} \subset \Gamma$ with

(i) $e^* \in \overline{\mathbb{T}\Gamma_{e^*}}^{w^*}$;

(ii) $\sup_{x^* \in \Gamma_{e^*}} \rho(x^*) < 1$;

(iii) for every $\varepsilon > 0$ and every nonempty relatively weak-star open $U \subset \Gamma_{e^*}$, there are a nonempty $V \subset U$, $x_1^* \in V$, $e \in S_X$, and $F \in \mathcal{L}(X)$ such that

(i)′ $\|F(e)\| = \|F\| = 1$;

(ii)′ $x_1^*(F(e)) = 1$;

(iii)′ $F^* x_1^* = x_1^*$;

(iv)′ with

$$V_1 = \{x^* \in \Gamma : \ \|F^* x^*\| + (1-\varepsilon)\|x^* - F^* x^*\| \le 1\},$$

one has $|x^*(F(e))| \le \rho(x_1^*)$ for every $x^* \in \Gamma \setminus V_1$;

(v)′ $\operatorname{dist}(F^* x^*, \operatorname{aco}(\{0\} \cup V)) < \varepsilon$ for every $x^* \in \Gamma$;

(vi)′ $|v^*(e) - 1| \le \varepsilon$ for every $v^* \in V$.

Here $\mathbb{T} = \{-1, 1\}$ in the real case.

We use only the standard fact that property β implies quasi-ACK structure; in fact it gives ACK_ρ structure for a fixed $\rho < 1$ [1, 2].

3 Audit of the supplied partial result

The conditional theorem in the supplied packet is correct. We record it in a slightly streamlined form.

Proposition 3.1 (Uniform active bound)**.** *Suppose that X_i has quasi-*ACK *data $(\Gamma_i, \rho_i, (\Gamma_{i,e^*})_{e^*})$, and put*

$$\Delta_i = \left(\bigcup_{e^* \in \operatorname{Ext}(B_{X_i^*})} \Gamma_{i,e^*} \right) \setminus \{0\}.$$

If

$$r := \sup_i \sup_{x^* \in \Delta_i} \rho_i(x^*) < 1,$$

then $(\bigoplus_i X_i)_{\ell_\infty}$ has ACK_r structure.

Proof. We first check the only nonstandard step. For one space X, let Δ be the corresponding active union. Removing 0 does not affect the weak-star approximation of a norm-one extreme point. Thus every $e^* \in \operatorname{Ext}(B_{X^*})$ lies in $\overline{\mathbb{T}\Delta}^{w^*}$. Krein–Milman gives

$$B_{X^*} = \overline{\operatorname{co}\operatorname{Ext}(B_{X^*})}^{w^*} \subset \overline{\operatorname{aco}(\Delta)}^{w^*} \subset B_{X^*}.$$

Consequently Δ is 1-norming.

Let U be a nonempty relatively weak-star open subset of Δ and pick $u^* \in U$. Some e_0^* satisfies $u^* \in \Gamma_{e_0^*}$. Then $U \cap \Gamma_{e_0^*}$ is nonempty and relatively weak-star open in $\Gamma_{e_0^*}$. Apply the local quasi-ACK axioms there. Restricting the ambient norming set from Γ to Δ preserves all six conditions, while condition (iv)′ improves to the constant estimate $\rho(x_1^*) \le r$. Hence X has ACK_r structure with norming set Δ.

Apply this to every X_i. The set

$$\Delta = \bigcup_i P_i^*(\Delta_i)$$

is 1-norming for the ℓ_∞-sum. Given a relatively weak-star open subset of Δ, choose one of its points and work in that coordinate. Extend the resulting vector and operator by zero in all other coordinates. For a functional supported in another coordinate, both its value on $F(e)$ and its image under F^* are zero, so conditions (iv)$'$ and (v)$'$ are immediate. The remaining conditions are exactly those in the chosen summand. This proves ACK$_r$ structure. □

The second paragraph is also an immediate application of Choi–Jung's Proposition 4.9 once the first paragraph has upgraded every summand to ACK$_r$. Thus the supplied result is valid; the missing uniformity is precisely where its proof cannot apply. The counterexample below has natural constants tending to one.

4 Regular-polygon summands

Choose strictly increasing integers $m_n \to \infty$, with $m_n \geq 4$, and put

$$h_n = \frac{\pi}{m_n}, \qquad u_{n,k} = (\cos(kh_n), \sin(kh_n)) \quad (0 \leq k < m_n).$$

On $\mathbb{R}^2$ define

$$\|x\|_n = \max_{0 \leq k < m_n} |\langle x, u_{n,k} \rangle|, \qquad X_n = (\mathbb{R}^2, \|\cdot\|_n). \tag{1}$$

We identify Euclidean vectors and Euclidean linear functionals.

Lemma 4.1. *For every n and $x \in \mathbb{R}^2$,*

$$\cos(h_n/2)|x|_2 \leq \|x\|_n \leq |x|_2. \tag{2}$$

Moreover, X_n has property β. One may use the pairs

$$(u_{n,k}, u_{n,k}) \qquad (0 \leq k < m_n)$$

and the parameter

$$\rho_n = \cos h_n < 1.$$

*In particular, every X_n has quasi-*ACK *structure, while $\rho_n \uparrow 1$.*

Proof. The lines spanned by the $u_{n,k}$ divide the projective unit circle into arcs of angular length h_n. A Euclidean unit vector is within angle $h_n/2$ of one of these lines, proving (2).

The polar of the unit ball in (1) is

$$B_{X_n^*} = \mathrm{co}\{\pm u_{n,k} : 0 \leq k < m_n\},$$

a regular $2m_n$-gon; in particular each $u_{n,k}$ has dual norm one. Also $\|u_{n,k}\|_n = 1$ and $\langle u_{n,k}, u_{n,k} \rangle = 1$. If $j \neq k$, then the angular separation of the corresponding unoriented lines is at least h_n, whence

$$|\langle u_{n,j}, u_{n,k} \rangle| \leq \cos h_n.$$

The definition of the norm says that the functionals are 1-norming. These are exactly the property-β requirements. □

Set

$$Z = \Big(\bigoplus_{n=1}^{\infty} X_n \Big)_{\ell_\infty}. \tag{3}$$

By (2), a member $z = (z_n)$ of Z is a uniformly Euclidean-bounded sequence.

5 Ultrafilter fibers and the extreme points of B_{Z^*}

Let $\beta\mathbb{N}$ be the Stone–Čech compactification of $\mathbb{N}$, viewed as the space of ultrafilters. For $p \in \beta\mathbb{N}$, define the fiber

$$X_p = \begin{cases} X_n, & p = n \text{ is principal,} \\ H := (\mathbb{R}^2, |\cdot|_2), & p \text{ is free.} \end{cases}$$

For $z = (z_n) \in Z$, let

$$Q_p z = \lim_{n \to p} z_n \in \mathbb{R}^2,$$

the coordinatewise ultralimit. If $p = n$, this is simply z_n.

Lemma 5.1. *For every $p \in \beta\mathbb{N}$, $Q_p : Z \to X_p$ is a norm-one quotient map, and*

$$\|Q_p z\|_{X_p} = \lim_{n \to p} \|z_n\|_n \qquad (z \in Z). \tag{4}$$

For a free p, the constant-sequence map

$$J : H \longrightarrow Z, \qquad Jx = (x, x, \ldots),$$

is an isometry and satisfies $Q_p J = I_H$.

Proof. The norms $\|\cdot\|_n$ converge uniformly to $|\cdot|_2$ on the Euclidean unit ball by (2). Since every $z \in Z$ is uniformly Euclidean bounded,

$$\big|\|z_n\|_n - |z_n|_2\big| \longrightarrow 0.$$

For a free p, continuity of the Euclidean norm therefore gives

$$\lim_{n \to p} \|z_n\|_n = \lim_{n \to p} |z_n|_2 = \left|\lim_{n \to p} z_n\right|_2.$$

The principal case is immediate, proving (4) and $\|Q_p\| \le 1$.

If $p = n$ is principal, the injection supported in coordinate n is an isometric right inverse. If p is free, then

$$\|Jx\| = \sup_n \|x\|_n = |x|_2,$$

because $\|x\|_n \le |x|_2$ and $\|x\|_n \to |x|_2$. Thus J is an isometric right inverse of Q_p. Hence every Q_p is a norm-one quotient. $\square$

The next proposition is the structural core of the argument.

Proposition 5.2 (Extreme-point formula)**.** *One has*

$$\operatorname{Ext}(B_{Z^*}) = \left\{ Q_p^* v^* : p \in \beta\mathbb{N},\ v^* \in \operatorname{Ext}(B_{X_p^*}) \right\}. \tag{5}$$

Proof. For $A \subset \mathbb{N}$, let P_A be the coordinate projection onto A. Since

$$Z = P_A Z \oplus_\infty P_{A^c} Z,$$

we have the canonical ℓ_1-decomposition

$$Z^* = (P_A Z)^* \oplus_1 (P_{A^c} Z)^*.$$

If $\phi \in \mathrm{Ext}(B_{Z^*})$, then for every A exactly one of the two components of ϕ is zero. Equivalently, ϕ is supported on exactly one of A and A^c, where "supported on A" means $\phi = \phi P_A$. The collection

$$p_\phi = \{A \subset \mathbb{N} : \phi = \phi P_A\}$$

is an ultrafilter: it chooses exactly one of A, A^c, is upward closed, and is closed under finite intersections.

Put $p = p_\phi$. If $Q_p z = 0$, then for every $\eta > 0$ the set

$$A_\eta = \{n : \|z_n\|_n < \eta\}$$

belongs to p, by (4). Since ϕ is supported on A_η,

$$|\phi(z)| = |\phi(P_{A_\eta} z)| \le \eta.$$

Thus ϕ annihilates $\ker Q_p$ and factors uniquely as $\phi = Q_p^* v^*$ for some $v^* \in B_{X_p^*}$. Since Q_p is a quotient, $\|v^*\| = \|\phi\| = 1$. If v^* were not extreme, its nontrivial midpoint decomposition would pull back to one of ϕ; hence v^* is extreme. This proves the inclusion from left to right in (5).

Conversely, let $v^* \in \mathrm{Ext}(B_{X_p^*})$ and suppose

$$Q_p^* v^* = \frac{\psi_1 + \psi_2}{2}, \qquad \psi_1, \psi_2 \in B_{Z^*}.$$

Fix $A \in p$ and decompose $\psi_j = \psi_{j,A} + \psi_{j,A^c}$. The average has norm one and is entirely supported on A. Therefore

$$1 \le \frac{\|\psi_{1,A}\| + \|\psi_{2,A}\|}{2} \le 1 - \frac{\|\psi_{1,A^c}\| + \|\psi_{2,A^c}\|}{2} \le 1.$$

It follows that both A^c-components vanish. Since this holds for every $A \in p$, the preceding kernel argument shows that each ψ_j factors through Q_p, say $\psi_j = Q_p^* v_j^*$ with $v_j^* \in B_{X_p^*}$. Extremality of v^* gives $v_1^* = v_2^* = v^*$, proving the converse. □

For a principal fiber, $\mathrm{Ext}(B_{X_n^*}) = \{\pm u_{n,k}\}$; for a free fiber, $\mathrm{Ext}(B_{H^*})$ is the Euclidean unit circle.

6 A localization inequality forced by quasi-ACK

The following consequence of the axioms is useful independently of the particular polygonal construction.

Lemma 6.1 (Localization inequality)**.** *Let X have a 1-norming set $\Gamma \subset B_{X^*}$ and a function $\rho : \Gamma \to [0, 1)$. Suppose that, for some $0 < \varepsilon < 1$, nonempty $V \subset \Gamma$, $x_1^* \in V$, $e \in S_X$, and $F \in \mathcal{L}(X)$, conditions* (i)′–(v)′ *of Definition 2.1 hold. Put*

$$y = F(e), \qquad D = \overline{\mathrm{aco}(J^* V)} \subset E^*,$$

where $J : E \to X$ is any contraction from a finite-dimensional Banach space. If $\psi \in \mathrm{Ext}(B_{X^*})$ and

$$a := \psi(y) > \rho(x_1^*),$$

then

$$\mathrm{dist}(J^*\psi, D) \le \varepsilon + \frac{1-a}{1-\varepsilon}. \tag{6}$$

Proof. Since Γ is 1-norming, $B_{X^*} = \overline{\mathrm{aco}(\Gamma)}^{w^*}$. Milman's converse to Krein–Milman yields a net $(\eta_\lambda) \subset \mathbb{T}\Gamma$ converging weak-star to ψ. Eventually $\eta_\lambda(y) > \rho(x_1^*)$. Write $\eta_\lambda = \theta_\lambda \gamma_\lambda$ with $\theta_\lambda \in \mathbb{T}$ and $\gamma_\lambda \in \Gamma$. Condition (iv)′ and its invariance under signs imply that $\gamma_\lambda \in V_1$, and hence

$$\|\eta_\lambda - F^*\eta_\lambda\| \le \frac{1 - \|F^*\eta_\lambda\|}{1-\varepsilon} \le \frac{1 - |\eta_\lambda(y)|}{1-\varepsilon}.$$

Condition (v)′ also remains valid after multiplication by a sign, since $\mathrm{aco}(\{0\} \cup V)$ is balanced. Applying the contraction J^* gives

$$\mathrm{dist}(J^*\eta_\lambda, D) < \varepsilon + \frac{1 - |\eta_\lambda(y)|}{1-\varepsilon}.$$

The space E^* is finite-dimensional, so weak-star convergence after restriction by J^* is norm convergence. Passing to the limit proves (6). □

Geometrically, equation (6) says that any extreme functional that nearly norms y must restrict close to the absolutely convex hull of the small local set V.

7 The counterexample

We now prove Theorem 1.1. Suppose, toward a contradiction, that the space Z in (3) has quasi-ACK data $(\Gamma, \rho, (\Gamma_{e^*})_{e^*})$.

Fix a free ultrafilter p and let $u_0 = (1, 0) \in H$. By Proposition 5.2,

$$\phi_0 = Q_p^* u_0 \in \mathrm{Ext}(B_{Z^*}).$$

Put $A = \Gamma_{\phi_0}$ and

$$r = \sup_{\gamma \in A} \rho(\gamma) < 1.$$

Since $\phi_0 \in \overline{\mathbb{T}A}^{w^*}$ and $\mathbb{T} = \{-1, 1\}$, either $\phi_0 \in \overline{A}^{w^*}$ or $-\phi_0 \in \overline{A}^{w^*}$. Choose $\sigma \in \{-1, 1\}$ so that, with

$$u = \sigma u_0, \qquad \phi = Q_p^* u,$$

we have $\phi \in \overline{A}^{w^*}$.

Choose an angle $\alpha \in (0, \pi/2)$ such that

$$\cos\alpha > r. \tag{7}$$

The strict inequality

$$\sin\alpha > 1 - \cos\alpha$$

holds for every $\alpha \in (0, \pi/2)$. By continuity, we may choose $\tau > 0$ so small that

$$0 < \tau < \min\{\alpha, 1/2\}, \qquad \alpha + 2\tau < \frac{\pi}{2}, \qquad \cos(\alpha + \tau) > r, \tag{8}$$

$$\sin(\alpha - \tau) - \tau > \tau + \frac{1 - \cos(\alpha + \tau)}{1 - \tau}. \tag{9}$$

Choose N so that $h_n < \tau$ for $n \ge N$. Next choose $0 < \delta < \tau$ with

$$\arccos(1 - \delta) < \tau, \tag{10}$$

and finally choose $0 < \varepsilon < \tau$.

Let $J : H \to Z$ be the constant-sequence isometry from Lemma 5.1, and define the tail vector

$$w_n^{(N)} = \begin{cases} 0, & n < N, \\ u, & n \geq N. \end{cases}$$

Because $u = \pm(1, 0)$, one has $\|u\|_n = 1$ for every n, so $w^{(N)} \in S_Z$ and $\phi(w^{(N)}) = 1$.

Consider the relatively weak-star open subset of A

$$U = \left\{ \gamma \in A : \|J^*\gamma - u\|_2 < \delta,\ \gamma(w^{(N)}) > 1 - \delta \right\}. \tag{11}$$

It is nonempty because $\phi \in \overline{A}^{w^*}$, while $J^*\phi = u$ and $\phi(w^{(N)}) = 1$. Apply the local quasi-ACK condition to U and ε. We obtain

$$\varnothing \neq V \subset U, \quad x_1^* \in V, \quad e \in S_Z, \quad F \in \mathcal{L}(Z).$$

Put $y = F(e)$. Then $\|y\| = 1$ and $x_1^*(y) = 1$.

The set

$$\mathcal{F}_y = \{\psi \in B_{Z^*} : \psi(y) = 1\}$$

is a weak-star compact face of B_{Z^*} containing x_1^*. Since $x_1^*(w^{(N)}) > 1 - \delta$, Krein–Milman applied to this face gives $\psi_0 \in \mathrm{Ext}(\mathcal{F}_y) \subset \mathrm{Ext}(B_{Z^*})$ such that

$$\psi_0(w^{(N)}) > 1 - \delta. \tag{12}$$

By Proposition 5.2, write

$$\psi_0 = Q_q^* v_0, \qquad v_0 \in \mathrm{Ext}(B_{X_q^*})$$

for some $q \in \beta\mathbb{N}$. Inequality (12) forces q to contain the cofinite tail $\{n \geq N\}$; hence q is free or it is a principal ultrafilter $n \geq N$. Moreover,

$$v_0(u) > 1 - \delta, \qquad t := Q_q y \in S_{X_q}, \qquad v_0(t) = 1. \tag{13}$$

Every extreme functional of every fiber has Euclidean norm one. Thus (10) and (13) imply that the Euclidean angle between v_0 and u is less than τ.

We next find another extreme functional in the same fiber. If q is free, then $X_q = H$, and $v_0(t) = 1$ forces $t = v_0$. Rotate v_0 through angle α in the direction away from u, obtaining a Euclidean unit functional v_1. Then

$$v_1(t) = \cos\alpha \geq \cos(\alpha + \tau), \qquad \alpha - \tau \leq \angle(v_1, u) \leq \alpha + 2\tau.$$

Suppose instead that $q = n \geq N$ is principal. Rotate coordinates so that $v_0 = (1, 0)$. Since $v_0(t) = 1$, write $t = (1, s)$. The two adjacent extreme functionals $(\cos h_n, \pm \sin h_n)$ both take value at most one on t; hence

$$|s| \leq \frac{1 - \cos h_n}{\sin h_n} = \tan(h_n/2). \tag{14}$$

Let $k = \lceil \alpha/h_n \rceil$ and $\theta = kh_n$. Then

$$\alpha \leq \theta < \alpha + h_n < \alpha + \tau.$$

Choose the extreme functional at angular displacement θ from v_0 in the sign of s (either sign if $s = 0$). Calling it v_1, we get

$$v_1(t) = \cos\theta + |s|\sin\theta \geq \cos\theta \geq \cos(\alpha + \tau). \tag{15}$$

Since $\angle(v_0, u) < \tau$, the triangle inequality for these small angles gives

$$\alpha - \tau \leq \angle(v_1, u) \leq \alpha + 2\tau < \frac{\pi}{2}. \tag{16}$$

The same two conclusions, (15) and (16), therefore hold in both the free and principal cases.

Set

$$\psi_1 = Q_q^* v_1.$$

By Proposition 5.2, $\psi_1 \in \mathrm{Ext}(B_{Z^*})$, and

$$a := \psi_1(y) = v_1(t) \geq \cos(\alpha + \tau) > r. \tag{17}$$

Also $J^*\psi_1 = v_1$.

Let

$$D = \overline{\mathrm{aco}(J^*V)} \subset H^* \cong \mathbb{R}^2.$$

Every $v^* \in V$ belongs to U, so $\|J^*v^* - u\|_2 < \delta$. It follows that

$$D \subset [-u, u] + \delta B_2. \tag{18}$$

By (16), the Euclidean distance from v_1 to the segment $[-u, u]$ equals $\sin\angle(v_1, u)$ and is at least $\sin(\alpha - \tau)$. Hence

$$\mathrm{dist}(v_1, D) \geq \sin(\alpha - \tau) - \delta > \sin(\alpha - \tau) - \tau. \tag{19}$$

On the other hand, $\rho(x_1^*) \leq r$, so (17) allows us to apply Lemma 6.1 to ψ_1. Using $\varepsilon < \tau$ and $a \geq \cos(\alpha + \tau)$ gives

$$\mathrm{dist}(v_1, D) \leq \varepsilon + \frac{1 - a}{1 - \varepsilon} < \tau + \frac{1 - \cos(\alpha + \tau)}{1 - \tau}. \tag{20}$$

Inequalities (19), (20), and (9) contradict one another. Therefore Z does not have quasi-ACK structure. This proves the first assertion of Theorem 1.1.

Finally, every X_n has property β by Lemma 4.1, hence also property quasi-β. If Z had property quasi-β, it would have quasi-ACK structure by Choi–Jung's Remark 2.2(b), contrary to what we have just proved. Thus property quasi-β is not stable under arbitrary ℓ_∞-sums either. □

8 Interpretation

The counterexample pinpoints the role of uniformity in Proposition 3.1. Each polygonal summand has an ordinary ACK_{ρ_n} structure, but

$$\rho_n = \cos(\pi/m_n) \longrightarrow 1.$$

At a free ultrafilter, the varying polygonal norms converge to the Euclidean norm. The local quasi-ACK set V is forced to cluster near a single Euclidean functional u, so its absolute convex hull restricts close to the line segment $[-u, u]$. Yet the norm-one face selected by the local operator contains an extreme fiber functional v_0 near u, and round geometry provides a second extreme functional v_1 at a controlled angle that still nearly norms the same vector. The localization inequality then says that v_1 must be close to $[-u, u]$, while elementary Euclidean geometry says it is uniformly far away.

This does not conflict with the positive result that $\ell_\infty(E)$ has quasi-ACK structure when one fixed finite-dimensional space E has quasi-ACK structure. Here the summands vary, and their polygonal geometry converges to a round fiber at infinity.

9 Literature check

The source question appears in the 2023 article of Choi and Jung [2]. Targeted searches through June 23, 2026, using the exact terms "quasi-ACK", "property quasi-beta", and "ell-infinity sum", found no published proof or counterexample resolving Remarks 3.4 or 4.8. A 2025 preprint on range strongly exposing operators has a section using quasi-ACK spaces but does not address this sum-stability question [4]. Thus the counterexample above was developed rather than taken from a located source.

References

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